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Contents

INTRODUCTION 2

PROBLEM FORMULATION AND REQUIREMENTS BLUEPRINT 4

1.0 Justification for a Data Warehouse 4

2.0 Type and Level of the Decision Making 8

3.0 Requirements of the Data Warehouse 13

3.1 Major Components of the Data Warehouse 18

4.0 Recommended Approach of Development: Inmon versus Kimball 22

REFERENCES 24

INTRODUCTION

The University of Botswana (UB) was established in 1982 as the first institution of higher learning in the country and today enrols in the region of 15,000 students across campuses in Gaborone, Francistown and Maun (University of Botswana, 2026). The University is organised into several faculties and departments, supported by administrative units including the Department of Academic Services, the Human Resources Division, the Bursary, Library and Information Services, and the Directorate of Information Technology (DIT), each of which generates and maintains its own operational data. Student information — admissions, registration, course enrolment, assessment and examination results, and graduation records — is captured and maintained through the University's online Student Administration System, which students and staff use every semester. In parallel, staff records are maintained by the Human Resources Division, financial and fee records by the Bursary, book and journal usage by the library, and research output by individual faculties and the research and innovation office. These systems were introduced at different times, for different purposes, and were not designed to share data with one another.

This fragmentation is not unique to UB. Kabakchieva (2015) observes that higher education institutions worldwide hold large volumes of student data in operational systems that are poorly structured for analysis, and that competition between institutions has driven the adoption of business intelligence tools to extract value from it. Pesonen, Fomkin and Jokipii (2018) describe the equivalent solution at Aalto University, whose data warehouse consolidates the student information system, research information system, financial systems and human resources system into a single analytical platform serving ad hoc analysis, dashboards and formal reporting. Within the region, Scholtz, Calitz and Haupt (2018) develop a business intelligence framework for institutional information management in a South African university, indicating that the problem and its solution are already recognised among UB's immediate peers.

As a result of the fragmentation at UB, when management - the Vice-Chancellor's office, Deans of faculties, the Registrar, the Bursar or the Director of Human Resources - needs a consolidated view, for example of enrolment trends across faculties and campuses, staff workload against student numbers, or fee arrears against sponsorship categories, this currently requires manually pulling extracts from each system and reconciling them, which is slow and prone to inconsistency.

A data warehouse -defined by Inmon (2005) as a subject-oriented, integrated, time-variant and non-volatile collection of data in support of management's decision-making process - would allow the University to bring together data from its Student Administration System, Human Resources system, Bursary/finance system, Library

system and faculty records into a single, consistent source that management and academic departments can query directly. This document constitutes Part A of the design: the problem formulation and requirements analysis blueprint.

PROBLEM FORMULATION AND REQUIREMENTS BLUEPRINT

1.0 Justification for a Data Warehouse

1.1 Does the University need a data warehouse?

Does the University really need a data warehouse?

The University currently operates several separate operational systems - the Student Administration System, the Human Resources system, the Bursary/finance system, and the library system - each holding data relevant to only one function. Faculties and administrative departments that need a cross-cutting view, for instance relating registration numbers to fee payment status or to staff establishment, must request extracts from each system separately and combine them manually. This is time-consuming and delays reporting to Senate, Council and the Ministry responsible for tertiary education.

Chaudhuri and Dayal (1997) identify precisely this condition — decision-support queries that span multiple heterogeneous operational databases — as the classic justification for a separate warehouse, since running such queries directly against transactional systems both performs poorly and degrades the operational workload. A data warehouse would let the University's management and administrative staff access consistent, integrated information from a single source without visiting each department, and would preserve more than four decades of student, staff and financial records in a form that supports trend analysis (Ponniah, 2010).

The answer is therefore yes. The University needs a data warehouse.

1.2 Is the University ready for one?

Ponniah (2010) proposes a readiness test based on observable organisational symptoms rather than on technology. Assessed against those indicators, several signs suggest the University is ready.

•Management waits for extended periods to receive consolidated reports, because each administrative system produces its own reports independently and combining them requires manual effort from IT and administrative staff.

•The University holds a large volume of data that is not structured to support decision-making, because it resides in separate, unintegrated systems.

•Some processes, particularly in smaller departments and at the Francistown and Maun campuses, still rely on spreadsheets and paper-based records that are later re-entered into central systems, introducing delay and risk of error.

•Different departments sometimes report slightly different figures for the same period, indicating the absence of a single version of the truth.

Watson and Wixom (2007) note that organisational and data-quality readiness, rather than tool selection, is what most often determines whether a warehouse initiative succeeds. On that measure the University is ready in terms of need and data availability but will require deliberate attention to data governance and to standardising codes across source systems before loading begins.

1.3 What is the current infrastructure?

The Student Administration System is the University's primary system of record for students. It is used for online registration each semester, for maintaining admissions and enrolment records, for scheduling lectures and examinations, and for recording assessment and examination results. This system is managed by the Department of Academic Services, which is responsible for student recruitment, admissions, enrolment management, registration, examinations administration and graduation.

Separately, the Human Resources Division maintains staff records - appointments, qualifications, leave and performance - in its own system, while the Bursary maintains student fee accounts, sponsorships and staff payroll in a finance system. The library maintains a system for cataloguing and circulation of books, journals and electronic resources, and individual faculties keep their own records of research output, postgraduate supervision and publications, often in spreadsheets rather than a central system.

These systems are disparate and not interlinked. A Dean wanting to relate a faculty's enrolment numbers to its staff establishment, or the Bursary wanting to relate outstanding fees to sponsorship type, must request extracts from more than one system and reconcile them manually.

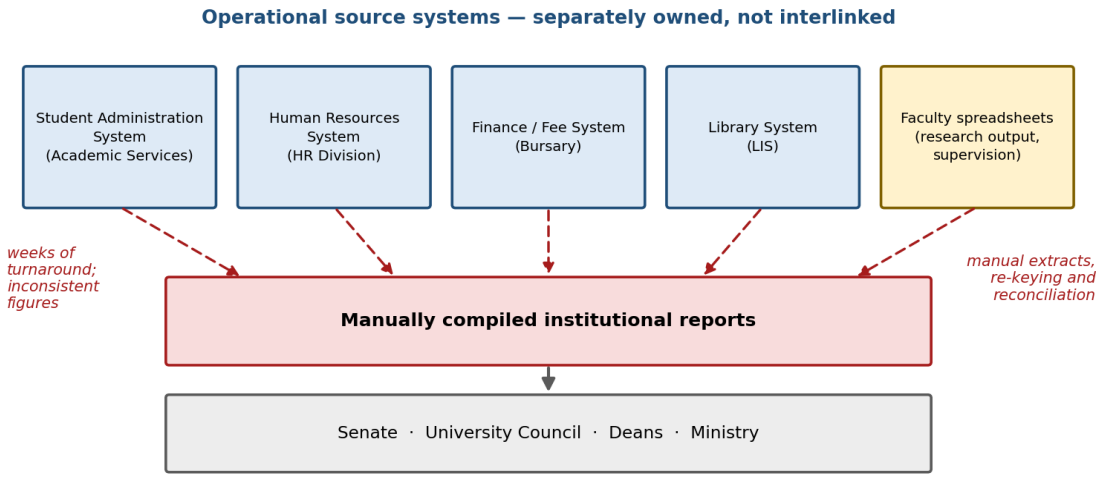


Figure 1:current as-is landscape for university of Botswana

Data captured across these systems for a single student over their time at the University typically includes personal and biographical details; admission and programme/faculty details; semester registration and course enrolment records; assessment and examination results; fee and sponsorship records; and library usage and, for postgraduates, research and supervision records. A properly designed data warehouse would draw this data together from each source without disrupting the operational systems themselves.

1.4 What would be the application area for the data warehouse?

The application area is best characterised as ALL — that is, an institution-wide scope rather than a single functional area. The warehouse would serve Academic Services and the Registrar's office, the Bursary/finance function, Human Resources, the Library, the Faculties, and the Research and Innovation office, each drawing on the same integrated source of data for planning and reporting relevant to its area, while the Vice-Chancellor's office and Council draw on all of these areas together for institution-wide decision-making. This mirrors the scope adopted by the University of Illinois System, whose enterprise data warehouse combines finance, human resources and student data in a single central source for reporting and analysis across the institution (University of Illinois System, 2026a).

1.5 Who will be using the data warehouse?

•The Vice-Chancellor's office and University Council, to obtain consolidated, institution-wide reports on enrolment, graduation and financial performance, supporting strategic planning and reporting to the Ministry responsible for tertiary education.

•Deans of Faculty, to identify trends within their faculty. A report showing declining enrolment or pass rates in a particular programme would help a dean decide where to focus resources or curriculum review.

•The Registrar and the Department of Academic Services, who need summarised reports on admissions, registration and examination patterns across faculties and campuses in order to plan lecture venues, examination timetables and staffing of academic support services, and to measure the efficiency of registration and graduation processes.

•The Bursar, to relate fee payment and sponsorship data to enrolment, helping forecast revenue and identify students at risk of deregistration for non-payment.

•The Director of Human Resources, to relate staff establishment and qualifications to student numbers by faculty, supporting staffing decisions.

•The Ministry responsible for tertiary education, as an external consumer of summarised, reliable statistics on enrolment, graduation and staffing when reviewing the University's performance and funding.

Each of these user groups should be presented with a view of the warehouse that provides timely and relevant information for their role, with access governed by role-based security. The University of Illinois model is instructive here: access to its warehouse is granted on request through designated unit security contacts, and authorised staff may then build custom reports and queries using a supplied business intelligence tool or any ODBC-compliant tool of their choice (University of Illinois System, 2026b).

1.6 What are the specific user needs and goals of the data warehouse?

The University requires a system that gives management and academic departments visibility of current performance and historical trends, and that adapts as the institution grows across its three campuses. Specifically, the system is required to:

•Produce consistent, institution-wide reports for Senate, Council and the Ministry.

•Collate data from the Student Administration System, HR, Bursary and Library while ensuring accuracy and consistency.

•Automate the data-collating process currently performed manually by administrative and IT staff.

•Scale as student numbers and the number of source systems grow, without adding major overhead.

•Integrate historical student, staff and financial data spanning the University's history since 1982.

•Support cross-functional analysis, for example relating enrolment to staffing and to fee collection.

•Automate the production and distribution of reports to faculties and management via email or the web.

•Provide easier, quicker and more comprehensive reporting and querying capabilities for Deans, the Registrar and the Bursar.

2.0 Type and Level of the Decision Making

2.1 What type and level of decisions will the warehouse support?

Decisions within an organisation are conventionally classified into three levels- strategic, tactical (management control) and operational -following the framework established by Anthony (1965) and extended to information systems by Gorry and Scott Morton (1971). The three levels differ in time horizon, in the degree of summarisation required, and in how structured the decisions are. A data warehouse is primarily an instrument of the strategic and tactical levels, because these are the levels at which integrated, historical and summarised data is needed (Turban, Sharda & Delen, 2011).

Figure 2 maps the University's decision makers onto these three levels.

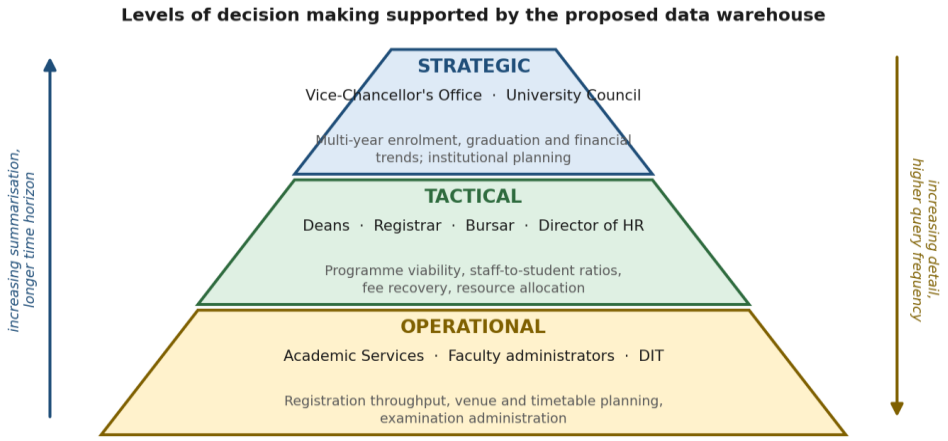


Figure 2: Levels of decision making supported by the proposed data warehouse

At the strategic level, the Vice-Chancellor's office and Council make largely unstructured decisions with a multi-year horizon — whether to open a programme, how to position the University against regional competitors, what to submit to the Ministry in support of funding. These decisions require heavily summarised data drawn from every subject area at once.

At the tactical level, Deans, the Registrar, the Bursar and the Director of Human Resources make semi-structured decisions on an annual or semester cycle: programme viability, staff-to-student ratios, fee recovery strategy, allocation of venues

and resources. This is the level at which the majority of warehouse queries will originate.

At the operational level, Academic Services staff, faculty administrators and DIT make structured, short-horizon decisions. The warehouse supports these only indirectly, through summarised process-efficiency measures such as registration throughput; day-to-day operational processing remains the responsibility of the source systems themselves.

2.2 What are the expectations?

A data warehouse is expected to help the University consolidate information that currently sits in separate systems, so that management no longer needs to wait for manually compiled reports. Because the warehouse is physically separate from the operational Student Administration, HR and finance systems, reports can be generated from it without slowing down day-to-day registration, payroll or fee processing -a separation of analytical from transactional workload that Chaudhuri and Dayal (1997) identify as one of the central architectural motivations for warehousing. By drawing on data from all these sources, Deans, the Registrar and the Bursar would no longer need to make decisions based on partial information from a single system, and the University could over time build towards predictive analysis of factors affecting student retention and performance, as Aalto University has done in using registry data held in its warehouse to model graduation probability and time-to-degree (Pesonen et al., 2018).

2.3 What are the criteria for assessing the value expected from the data warehouse?

Expectations must be expressed as measurable criteria if the investment is to be evaluated after implementation. Watson and Wixom (2007) argue that the value of business intelligence investments should be assessed against defined operational and decision-quality measures rather than asserted in general terms. Table 1 sets out the criteria proposed for this project, together with the current baseline and the target to be achieved within twenty-four months of go-live.

Table 1 Criteria for assessing the value expected from the data warehouse

Criterion	Measure	Baseline and target
Reporting turnaround	Elapsed time to produce a consolidated institutional report for Senate or Council	Baseline: 2-4 weeks of manual compilation. Target: same working day from the warehouse.
Manual effort	Staff hours per reporting cycle spent extracting and reconciling data	Target: reduction of at least 70% in administrative and DIT hours spent on routine reporting.
Single version of the truth	Number of discrepancies between figures reported by different departments for the same period	Baseline: recurring discrepancies. Target: zero unexplained discrepancies on core enrolment, staffing and fee measures.
Integration coverage	Number of source systems loaded into the warehouse	Target: four core systems (SAS, HR, Bursary, Library) within 24 months; faculty research records thereafter.
Historical depth	Years of comparable history available for trend analysis	Target: a minimum of ten years of conformed history at go-live.
User adoption	Proportion of Deans and heads of administrative units querying the warehouse directly each semester	Target: at least 80% of eligible users active each semester.
Decision quality	Proportion of Senate and Council papers sourced directly from the warehouse without manual reconciliation	Target: at least 90% of routine institutional papers.
New analytical capability	Availability of analyses not previously possible	Target: retention and fee-risk models operational by the end of year two.

2.4 Will it help executives and managers to do better planning and make better decisions?

At management level, information is often required quickly -ahead of a Senate or Council meeting, for instance - and there is little time to wait weeks for departments to compile figures manually. Because the data warehouse would be separate from the operational systems, faster reporting would not affect the performance of registration or payroll processing. A consolidated view of historical activity would allow the Vice-Chancellor's office and Deans to see which programmes, faculties or campuses have grown or declined over time, and to adjust recruitment, staffing and resource allocation accordingly. Kabakchieva (2015) demonstrates this in practice, showing how a business intelligence layer over student data allows institutional managers to monitor performance indicators continuously rather than retrospectively.

2.5 Is it going to improve the bottom line? How?

For a publicly funded university the bottom line is best understood as the efficiency of the institution's cost base and the security of its revenue, rather than as profit. The warehouse would improve both.

•Reduced cost of reporting: Allowing quick access to data from all four core systems through one platform removes the need for staff in each department to prepare extracts separately, reducing the administrative cost of compiling routine reports and freeing IT and administrative staff for other work.

•Improved fee recovery: Relating fee payment and sponsorship data to enrolment allows the Bursary to identify students at risk of deregistration for non-payment early enough to intervene, protecting tuition revenue that would otherwise be written off.

•Better resource allocation: Accurate staff-to-student ratios by faculty allow staffing and venue decisions to be made on evidence rather than on estimates, avoiding both over-provision and the cost of remedial hiring.

•Reduced rework: Figures drawn from a single integrated source replace the current situation in which different departments report slightly different numbers for the same period, eliminating the reconciliation work that discrepancy currently generates.

2.6 Is it going to increase market share? If so, by how much?

A goal common to all universities is to make better strategic decisions than their peers, particularly other institutions in the region such as BIUST and the private universities competing for students and research funding. In a higher education context, market share is most usefully defined as the University's share of national tertiary enrolment and of postgraduate intake.

It should be stated plainly that a data warehouse does not itself recruit students; it is an enabler rather than a direct cause of growth. Any target must therefore be modest and must be attributed to the decisions the warehouse makes possible rather than to the technology. On that basis the project proposes a target of a two to three percentage point gain in the University's share of national tertiary enrolment over five years, achieved through three specific mechanisms:

1.Earlier identification of declining programmes, allowing curriculum review or closure before enrolment losses compound, and earlier identification of growing programmes, allowing capacity to be added while demand exists.

2.Improved student retention. Because a substantial part of enrolment loss is attrition rather than failure to recruit, retention modelling of the kind piloted at Aalto (Pesonen et al., 2018) converts directly into retained enrolment share.

3.Faster and more credible submissions to the Ministry and to accreditation bodies, supporting the funding and approvals on which programme expansion depends.

The realism of this target should be reviewed against actual baseline figures once the first year of conformed history is loaded, and it is recommended that it be treated as provisional until then.

2.7 What are the risks faced without the benefits of a data warehouse, and what losses are likely to be incurred?

Without a data warehouse, the University would continue to rely on manually compiled, single-system reports, and would risk falling behind other universities that have already invested in integrated institutional reporting - a trend that tends to encourage further adoption elsewhere as institutions compete for students, staff and research funding (Kabakchieva, 2015). Specifically, the University would forgo three major advantages and incur the corresponding losses.

4.Integrating data from multiple sources. The University has several systems that output data in different formats and are not linked. The loss is the continuing cost of manual reconciliation and the reputational cost of inconsistent institutional statistics presented to Council and to the Ministry.

5.Performing new types of analysis, such as identifying factors associated with student retention and graduation. The loss is the tuition revenue and the graduate output associated with students who withdraw and who could have been identified and supported earlier.

6.Reducing the cost of accessing historical data. The University holds more than four decades of records that are costly to consolidate manually each time they are needed. The loss is both the recurring staff cost, and the opportunity cost of decisions taken without the benefit of trend evidence because assembling it was too slow.

A fourth risk is worth noting: the risk of decisions actively taken on wrong figures. Where departments report differing numbers for the same period and no authoritative source exists to arbitrate, there is no guarantee that the figure reaching Council is the correct one.

3.0 Requirements of the Data Warehouse

3.0.1 What software methodology is employed to solicit the requirements, and why?

The requirement-gathering process was conducted through structured interviews with staff in the Department of Academic Services and the Directorate of Information Technology, who between them are responsible for the Student Administration System and for institutional reporting.

Structured interviews were selected for three reasons. First, the population of stakeholders is small, senior and geographically dispersed across three campuses; interviews reach them individually without requiring the scheduling commitment that a workshop-based technique such as Joint Application Development would demand. Second, a fixed question set can be put consistently to staff responsible for different source systems, which makes their responses directly comparable and therefore easier to consolidate into a single set of requirements. Third, as Sommerville (2016) notes, interviewing is effective at eliciting the domain knowledge and organisational context that stakeholders hold tacitly but is weaker at surfacing requirements stakeholders have not thought to articulate.

To compensate for that weakness, structured interviews were supplemented by document analysis of existing institutional reports and Senate and Council papers, which reveals what management actually asks for in practice rather than what stakeholders report asking for. The combination of the two techniques is the methodology adopted for this project.

3.0.2 What are the specific business requirements?

Business requirements describe what the institution must be able to do.

Table 2: Business requirements mapped to objectives

Ref	Business Requirement	Objectives served
BR1	Produce a single consolidated institutional report covering enrolment, staffing and finance for Senate, Council and the Ministry	Establish a single version of the truth for institutional reporting
BR2	Allow Deans to analyse enrolment, pass rates and graduation outcomes by programme, faculty, campus and year	Support tactical decisions on programme viability and curriculum review

BR3	Allow the Bursar to relate fee payment and sponsorship status to enrolment and to identify students at risk of deregistration	Protect tuition revenue and support financial forecasting
BR4	Allow HR to relate staff establishment and qualifications to student numbers by faculty Support	evidence-based staffing and workload decisions
BR5	Allow the Registrar to measure registration, examination and graduation process efficiency across campuses	Improve operational efficiency of academic administration
BR6	Provide role-based access so that each stakeholder sees data relevant and permitted to their role	Protect personal data and maintain confidentiality
BR7	Automate scheduled production and distribution of standard reports to faculties and management	Remove manual effort from routine reporting
BR8	Support ad hoc querying without requiring programming skills	Reduce dependence on DIT for routine information requests

3.0.3 What are the specific data requirements?

Data requirements describe what data the warehouse must hold, at what level of detail, and how often it must be refreshed. Kimball and Ross (2013) stress that declaring the grain, the level of detail represented by a single row, is the decisive design step, since every dimension and measure follows from it. Table 3 states the grain, history and refresh requirements for each subject area.

Subject area	Grain and key data	History and refresh
Student enrolment	One row per student per course per semester. Student ID, programme, faculty, campus, mode of study, sponsorship category, registration status.	Minimum 10 years of history at go-live, extending towards 1982 where records permit. Nightly refresh during registration, weekly thereafter
Academic performance	One row per student per course per assessment period. Continuous assessment and examination marks, final grade, pass/fail outcome, credits earned.	Minimum 10 years. Refreshed at the close of each examination cycle. Graduation One row per student per award. Award type, classification, faculty, date conferred, time-to-degree. Full history where records permit. Refreshed annually after each graduation.
Human resources	One row per staff member per establishment period. Post, grade, faculty or unit, qualification, appointment and separation dates, full-time equivalent	Minimum 10 years. Monthly refresh.
Finance and fees	One row per student per fee transaction. Amount billed, amount paid, outstanding balance, sponsorship source, payment date	Minimum 10 years. Nightly refresh.
Library usage	One row per loan or electronic resource access event, aggregated to faculty and programme level.	Minimum 5 years. Weekly refresh
Research output	One row per publication or supervision record. Author, faculty, output type, year, postgraduate supervised.	Full history where records permit. Annual refresh, subject to remediation (see Table 5).

Table 3: Data requirements by subject area

Conformed dimensions must be shared across all subject areas so that the same faculty, programme, campus and time dimensions mean the same thing in every data mart (Kimball & Ross, 2013). Without this, a query relating enrolment to staffing would produce meaningless results. Estimated initial volume, based on an enrolment of approximately 15,000 students and ten years of history, is in the order of tens of millions of fact rows - a volume well within the capability of a conventional relational warehouse platform and not requiring specialised big-data infrastructure.

3.0.4 What type of data warehouse is to be built, and where will it be kept?

An Enterprise Data Warehouse (EDW) is recommended: a centralised warehouse providing decision-support services across the whole institution, organised and classified by subject area -students, staff, finance, library and research (Ponniah, 2010). An EDW best suits the University of Botswana so that data from all three campuses can be stored and accessed from one place, and so that the conformed dimensions described above are enforced centrally rather than negotiated between departments.

The data warehouse would be kept and maintained at the Directorate of Information Technology in Gaborone, which already hosts the University's central systems and data centre. Locating it there places it under existing security, backup and network management arrangements, and keeps it physically close to the source systems from which it will be loaded, minimising extraction latency. An on-premises deployment is recommended in preference to cloud hosting at this stage, given that student and staff records constitute personal data whose custody and cross-border transfer would require separate legal consideration.

3.0.5 Where will the data come from?

Table 4: Source system to subject area matrix

Source system	Owner	Subject areas populated
Student Administration System	Department of Academic Services	Student enrolment, academic performance, graduation (BR1, BR2, BR5)
Human Resources system	Human Resources Division	Human resources, staff establishment (BR1, BR4)
Bursary / finance system	Bursary	Finance and fees, sponsorship (BR1, BR3)
Library system	Department of Library and Information Services	Library and resource usage (BR2)
Faculty and departmental spreadsheets	Individual faculties, Research and Innovation office	Research output, postgraduate supervision (BR2)

3.0.6 Have we all the needed data?

For the most part, yes. The University already holds the data required for the enrolment, performance, staffing and finance subject areas, collected and maintained as part of normal daily operations, and it is therefore well suited to being drawn into a warehouse without requiring new data-collection processes. However, an honest assessment identifies four gaps that must be addressed before or during implementation.

Subject area	Gap identified	Proposed remediation
Research output and postgraduate supervision	Held locally in faculty spreadsheets with no common format or central custodian; completeness varies by faculty	Defer to a later phase. Define a standard template and a central custodian in the Research and Innovation office before attempting to load
Francistown and Maun campus records	Some processes still rely on spreadsheets and paper records re-entered centrally, introducing delay and transcription error	Audit affected processes; prioritise source-system capture at the point of origin before loading these records
Historical records prior to system computerisation	Early records may exist only on paper, limiting how far back conformed history can extend	Set the initial history horizon at ten years; treat earlier digitisation as a separate project
Reference and code consistency	Faculty and programme codes differ between the Student Administration System, HR and the Bursary	Establish a master reference data set and conform codes during the transformation stage(Section 2.4.2)

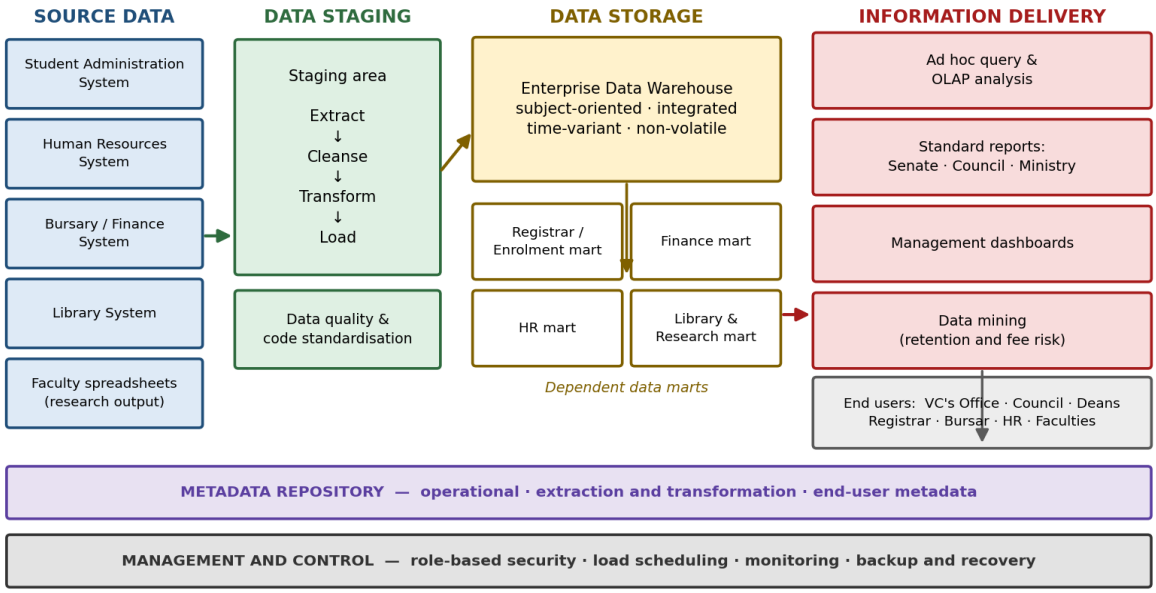
Table 5: Data availability assessment and identified gaps

Identifying these gaps at the requirements stage rather than discovering them during loading is itself a purpose of the blueprint. Watson and Wixom (2007) observe that data quality problems left unresolved at this point are the most common cause of warehouse projects failing to deliver their expected value.

3.1 Major Components of the Data Warehouse

Ponniah (2010) decomposes a data warehouse into a set of distinct architectural components, each with its own function and its own requirements. Figure 3 shows these components as they would be configured for the University of Botswana; each is then described in turn, together with the specific requirements it must satisfy in this environment.

Figure 3 : Major architectural components of the proposed data warehouse



3.1.1 Source data component

The source data component comprises the operational systems from which the warehouse draws. For the University these are the four production systems and the faculty spreadsheets identified in Table 4. Ponniah (2010) classifies warehouse source data as internal production data, archived data, internal private data held by individuals, and external data. All four categories are present here: production data in the Student Administration, HR, Bursary and Library systems; archived data in historical student and financial records; internal private data in the faculty spreadsheets; and external data in the form of Ministry and Statistics Botswana benchmarks against which institutional performance is compared.

The principal requirement of this component is non-intrusive access. Extraction must not degrade the performance of registration, payroll or fee processing, which means extraction windows must be scheduled outside peak operational hours and must use read-only connections.

3.1.2 Data staging component

Data staging extracts data from the University's different source systems, transforms it into a form useful for reporting, and loads it into the warehouse. There are two main approaches. With ETL, data is first extracted into a staging area and transformed into a consistent form before being loaded. With ELT, data is extracted and loaded first, with transformation carried out within the warehouse itself.

The University of Botswana would be best served by the ETL approach, because it allows student, staff and financial data drawn from the Student Administration System, HR and Bursary to be cleaned and standardised — for example, resolving the inconsistent faculty and programme codes identified in Table 5 — before it is loaded, preventing duplicate or inconsistent records from reaching the warehouse at all. Given that data-quality remediation is a known requirement in this environment rather than a hypothetical risk, transforming before loading is the safer choice. Figure 4 sets out the process.

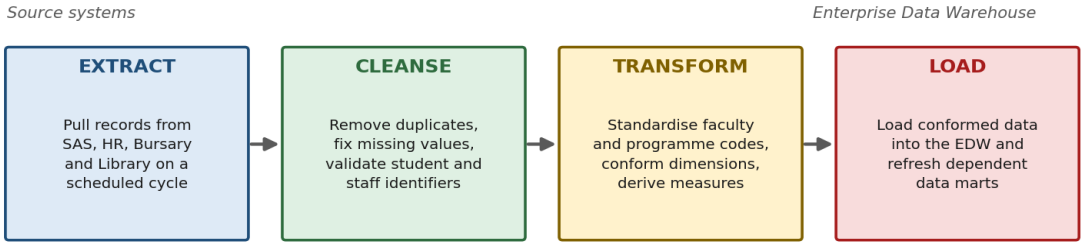


Figure 4: The ETL process applied to the University's source systems

Key requirements of the data staging component are therefore: the ability to connect to each of the University's existing source-system databases; a dedicated staging area physically separate from both source systems and the warehouse; support for scheduled, restart able loads so that a failure part way through does not corrupt the warehouse; data-quality rules covering duplicate detection, missing-value handling and referential integrity; and an audit trail recording what was loaded, when, and from where.

3.1.3 Data storage component

The data storage component holds the integrated data once staging is complete. It comprises the enterprise data warehouse itself and the dependent data marts derived from it. Following Inmon (2005), the warehouse is subject-oriented (organised around students, staff, finance, library and research rather than around applications), integrated (conformed to consistent codes and formats), time-variant (every record carries a time context so that history is preserved) and non-volatile (data is loaded and read but not updated in place, so that a figure reported last year remains reproducible this year).

Non-volatility deserves emphasis in a university context. When a Dean queries pass rates for a past academic year, the warehouse must return the same answer it returned when that year's report was first produced, even if the underlying operational record has since been corrected. Preserving that reproducibility is a requirement, not an incidental property.

Four dependent data marts are proposed- Registrar/enrolment, finance, human resources, and library and research -each drawing on the central warehouse and sharing its conformed dimensions. Storage requirements include capacity for at least ten years of history with headroom for growth, indexing and aggregate structures tuned for analytical rather than transactional access, and partitioning by academic year to keep query performance stable as history accumulates.

3.1.4 Information delivery component

The information delivery component is what users actually see, and its design determines whether the warehouse is adopted or ignored. Four delivery modes are required for the University, matched to the decision levels shown in Figure 2.

•Standard scheduled reports for Senate, Council and Ministry submissions, produced automatically on a defined calendar and distributed by email or the web, satisfying BR7.

•Management dashboards presenting key indicators — enrolment by faculty and campus, staff-to-student ratio, fee recovery rate, graduation rate — for the Vice-Chancellor's office and Deans, refreshed without manual intervention.

•Ad hoc query and OLAP analysis allowing Deans, the Registrar and the Bursar to drill down from institutional totals to faculty, programme and campus level without requesting help from DIT, satisfying BR8.

•Data mining for the predictive analyses identified earlier, particularly student retention and fee-payment risk modelling.

The University of Illinois System offers a working precedent for how this component is provisioned in practice: authorised staff may create custom reports, queries and data downloads against finance, human resources and student data using either a supplied business intelligence tool or any ODBC-compliant tool of their own choosing (University of Illinois System, 2026b). Adopting the same principle of tool flexibility would allow UB faculties to continue using the spreadsheet-based tools they already know while drawing on governed warehouse data underneath.

3.1.5 Metadata component

Metadata is data about the data, and it is the component most often neglected in student designs and in practice. Ponniah (2010) distinguishes three types, all of which are required here.

•Operational metadata records the structure, location and ownership of data in each source system - which field in the Student Administration System holds programme code, and who is responsible for it.

•Extraction and transformation metadata records how each data element was derived: which source it came from, what cleansing and transformation rules were applied, and when it was last loaded. Without this, a disputed figure cannot be traced back to its origin.

•End-user metadata provides the business meaning of each element in language Deans and administrators understand - what precisely counts as an enrolled student, whether a deferred registration is included, how full-time equivalent staff is calculated.

End-user metadata directly addresses the University's present difficulty of departments reporting different figures for the same period. In most cases the discrepancy arises not from arithmetic error but from different definitions applied to the same term. A published business glossary held as warehouse metadata resolves this at source.

3.1.6 Management and control component

The management and control component coordinates the other components and governs their operation. Its responsibilities are scheduling and monitoring load jobs and alerting DIT to failures; enforcing role-based security so that, for example, a dean sees full detail for their own faculty but only summary data for others, satisfying BR6; backup and recovery; and monitoring query performance and storage growth over time.

Security deserves weight here. The warehouse will concentrate personal data about students and staff -including academic performance and financial standing - in a single repository, which raises the consequence of unauthorised access well above that of any individual source system. Access must be granted by role through a defined authorisation process. The University of Illinois System's practice of routing all warehouse access requests through designated unit security contacts rather than accepting them directly from individuals offers a suitable model (University of Illinois System, 2026b).

4.0 Recommended Approach of Development: Inmon versus Kimball

This section reviews the two leading data warehouse development philosophies in relation to the University of Botswana scenario and recommends a suitable approach.

Inmon (2005) advocates a top-down approach: building a large, centralised enterprise data warehouse first, consolidating all available data from the Student Administration System, HR, Bursary and Library into a normalised, subject-oriented, integrated, time-variant and non-volatile repository that supports decision-making across the whole institution, and only then building departmental data marts from it. The architecture is consistent by construction, because every mart derives from one integrated source.

Kimball and Ross (2013) advocate a bottom-up approach, starting with mission-critical data marts that serve the analytic needs of individual business processes - for example a Registrar's mart focused on enrolment and registration - and integrating these marts through a shared set of conformed dimensions held in what they term the enterprise bus architecture. The approach delivers a working solution to one department relatively quickly before expanding to others.

Table 6 compares the two approaches against the criteria that matter for this project.

Table 6: Comparison of the Inmon and Kimball approaches

Criterion	Inmon (top-down)	Kimball (bottom-up)
Starting point	Centralised normalised enterprise warehouse covering all subject areas	A single dimensional data mart for one high-priority business process
Time to first deliverable	Long; little visible output until the full warehouse is built	Short; a working mart can serve the Registrar within months
Initial cost and risk	High up-front investment, with benefit realised late	Lower initial cost, benefit realised early, risk spread across phases
Consistency mechanism	Guaranteed by the single integrated repository	Achieved through conformed dimensions that must be defined and enforced deliberately
Skills required	Substantial data-modelling capacity in-house from the outset	Smaller initial team; capability can be built incrementally
Fit to UB's funding model	Difficult to justify a large single allocation in a constrained public budget	Suits phased funding, with each phase justified by demonstrated value
Principal weakness in this context	Risk of losing institutional support before any output is visible	Risk of inconsistent marts if conformed dimensions are not enforced

Recommendation

The recommendation for the University of Botswana is to adopt Kimball's bottom-up approach, beginning with a data mart for the Department of Academic Services covering enrolment, registration and examination data.

Three considerations support this. First, enrolment and registration reporting is the area in which management most urgently needs consolidated information, so the first deliverable addresses the most pressing need rather than the most convenient one. Second, the University operates within a constrained public budget in which a large single capital allocation is difficult to secure; a phased approach allows each stage to be funded on the strength of value already demonstrated. Third, DIT's existing warehousing capability would need to be built up in either case, and an incremental approach allows that capability to develop alongside the delivery rather than being required in full at the outset.

This recommendation is made subject to one firm condition. The principal risk of the Kimball approach is that independently built marts drift into inconsistency, reproducing at warehouse level exactly the fragmentation the project exists to remove. The conformed dimensions for faculty, programme, campus, student and time must therefore be defined and placed under central governance before the first mart is built, and every subsequent mart must be required to use them without local modification. On that condition, the Kimball approach delivers value soonest at acceptable risk.

It should be noted that the two philosophies are not mutually exclusive in current practice. Golfarelli and Rizzi (2009) observe that many organisations adopt hybrid architectures, and the phased path recommended here - conformed marts feeding towards an institution-wide warehouse - is in effect such a hybrid, taking Kimball's delivery sequence while retaining Inmon's ambition of a single integrated institutional repository as the end state.

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